

KEERTHI MEDHA MOTHUKURI

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Fourth-year **AI/ML undergraduate** with hands-on research and industry experience building production-grade AI systems. Currently conducting research on multimodal deep learning for content moderation at **IIT Guwahati**, and formerly engineered advanced RAG engines and AI agents during an industry internship at **Infosys Springboard**. Experienced in end-to-end AI development from model training to deploying live applications using **Streamlit** and **Flask**.

EXPERIENCE

Research AI/ML Intern June 2026 - Present

IIT Guwahati — Department of CSE

- **Implementing and developing** a multimodal machine learning pipeline using Hugging Face transformer models (BERT, VideoMAE, and Audio Spectrogram Transformer) to predict content moderation risks.
- **Designed and implemented** multimodal transformer pipelines for content moderation to automatically detect sensitive, harmful, or policy-violating content in real-time.
- **Testing, evaluating, and optimizing** deep learning models to improve performance, accuracy, and inference speed.
- **Utilizing GPU computing clusters** to train and evaluate models on large-scale video datasets.
- **Documented experiments**, model performance, and evaluation results.

AI Intern Sept 2025 - Nov 2025

Infosys Springboard

- **Architected a high-fidelity RAG engine** using OpenAI GPT-4 API, Pinecone vector database, and vector embeddings, achieving a 40% reduction in hallucination rates.
- **Engineered specialized AI agents** using GPT-4, prompt engineering, and Pinecone vector database for autonomous Legal Drafting and Case Summarization.
- **Delivered a production-ready application** featuring hybrid search (Vector + Keyword) and latency-optimized inference pipelines.
- **Collaborated with cross-functional teams** to document system architecture and API endpoints, improving code maintainability.

PROJECTS

Memora: A Cognitive Prosthetic for Dementia Care Jan 2025 - May 2025

- **Developed an "External Hippocampus"** for dementia patients to retain independence through personalized memory assistance.
- **Architected a Multi-Agent system** using Next.js, TypeScript, and Qdrant for hybrid RAG context retrieval.
- **Utilized multimodal inputs** to store and recall human-centric context via semantic vector search.

AI-Based English Proficiency Assessment & TOEFL/GRE Prediction Ongoing Project

- **Architecting a comprehensive ML framework** to assess linguistic proficiency and map it to TOEFL/GRE benchmark levels.
- **Developing an NLP pipeline** for feature extraction including vocabulary richness, grammatical accuracy, and syntactic complexity.
- **Leveraging supervised learning models** for multi-class proficiency categorization and optimizing classification reliability.

EDUCATION

Bachelor of Technology — Artificial Intelligence & Machine Learning Expected 2027

Vasireddy Venkatadri Institute of Technology (VVIT) — Department of Computer Science GPA: 8.34/10

SKILLS

Generative AI & LLMs: GPT-4, Prompt Engineering, RAG, LangChain, Pinecone, Hugging Face Transformers.

Programming: Python, Java, C, HTML/CSS.

Machine Learning & Natural Language Processing (NLP): TensorFlow, scikit-learn, Computer Vision (YOLO), Speech Recognition.

Databases & Cloud: Pinecone, SQL, AWS.

Soft Skills: Communication, Team Collaboration, Technical Documentation, Problem Solving.

CERTIFICATIONS

Cloud Foundations (AWS Academy) – 2026

Generative AI Professional (Oracle) – 2025

Large Language Models (NPTEL) – 2025